What is SQL injection? Describe two threats that can occur if the Blackboard website has SQL injection vulnerabilities.

A security flaw known as SQL injection enables attackers to change the SQL queries that are sent to a database by "injecting" or introducing malicious code. This usually happens when an application or website uses untrusted input—such as data supplied in a form—into a SQL query without properly validating or sanitizing it. SQL injection can be used to gain illegal access to a database, which could reveal or change private data.

If the Blackboard website has SQL injection vulnerabilities, two major threats that could occur include:

1. **Unauthorized Data Access**: Sensitive data, including grades and student records, as well as private information like passwords and email addresses, could be recovered by attackers from the database. Identity theft, privacy violations, and improper use of personal data could result from this.
2. **Data Manipulation**: Attackers could change or remove information from the Blackboard system, changing assignments, grades, or course enrollments. Both students and teachers may experience pandemonium as a result, and the integrity of academic records may be compromised.

Describe an SQL injection tool and its use.

SQLmap is a well-known SQL injection tool. An open-source program called SQLmap makes it easier to find and take advantage of SQL injection flaws in online applications. Testers and security experts can use it to find SQL vulnerabilities and carry out many SQL injection techniques, including enumerating tables and columns, extracting database information, and even executing privilege escalation. Security researchers, penetration testers, and ethical hackers frequently use SQLmap to evaluate online applications' security and make sure they are safe from SQL injection attacks. Unauthorized usage is prohibited, thus it's crucial to utilize these tools sensibly and with consent.

Describe three abuse cases that could occur on a login page for an online shopping website.

For a login page with use cases such as login, sign-up, and password recovery, potential abuse cases include:

1. **Credential Stuffing**: Assuming that some users have reused their passwords, attackers might attempt to enter into the shopping site using credentials they have stolen from other websites. Unauthorized account access may result from this, allowing hackers to browse or make transactions using the user's account.

2. **Account Enumeration**: In order to see if the system reveals which accounts exist, attackers could misuse the "Forgot Password" feature by entering alternate usernames or email addresses (for example, by giving different error messages for invalid vs. genuine accounts). Targeted assaults on verified accounts can then be carried out using this information.
3. **SQL Injection in Input Fields**: Attackers may try to get direct access to the database and obtain private data, including hashed passwords and all stored usernames, by entering malicious SQL code in password or username fields if appropriate input validation is not in place.